

EXPLORATORY DATA ANALYSIS USING POWER BI

WEEK 7 – STUDENT PERFORMANCE DATA CLEANING & TRANSFORMATION

Submitted by :

Divya Dharshni P

E25DA014

1. INTRODUCTION

This focuses on cleaning, transforming, and preparing the Student Performance dataset using **Power BI**. The raw data was examined for errors, duplicates, missing values, and inconsistent data. New calculated columns and measures were created to analyze students' performance effectively.

2. OBJECTIVES

- To import the Student Performance dataset into Power BI.
- To inspect and clean the dataset.
- To identify and handle duplicates, missing values, and errors.
- To standardize categorical values and data types.
- To create Total Score and Average Score columns.
- To classify students based on Performance Grade.
- To create Pass/Fail classification.
- To unpivot subject score columns.
- To create DAX measures for performance analysis.

3. TOOLS USED

- **Power BI Desktop**
- **Power Query Editor**
- **DAX (Data Analysis Expressions)**
- Student Performance in Exams Dataset

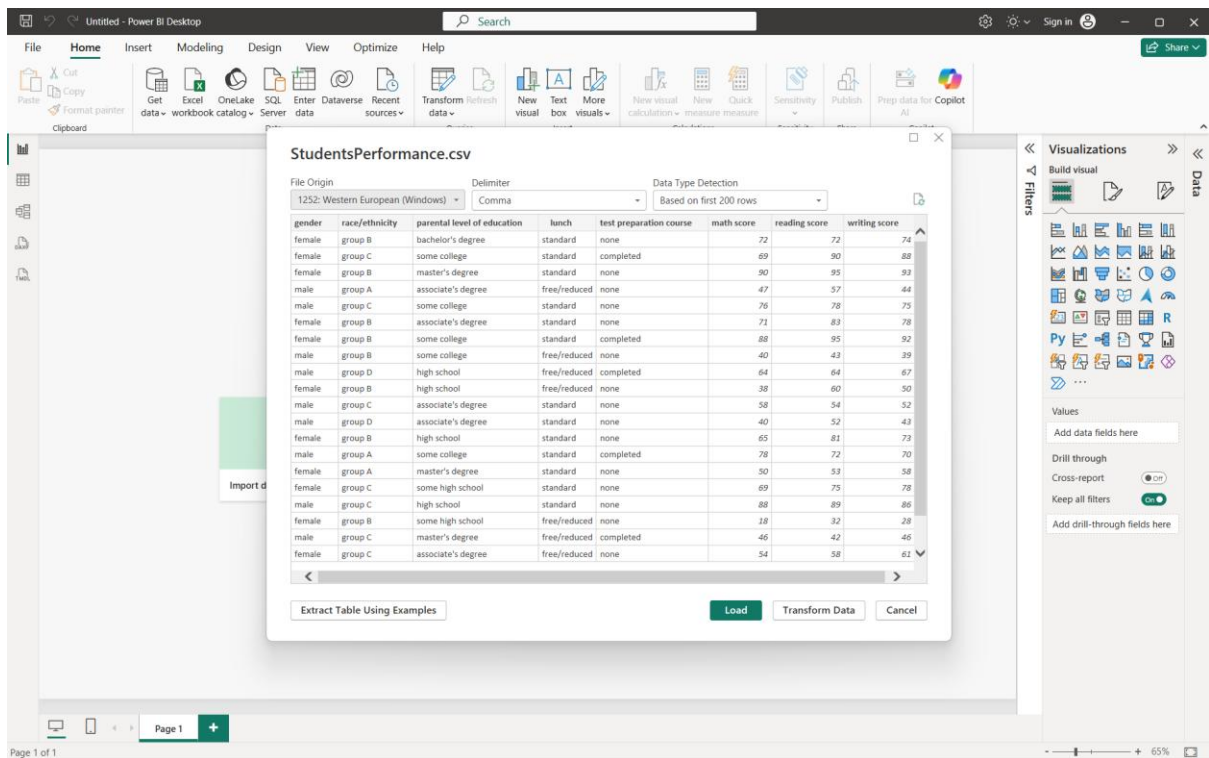
4. IMPORTING THE DATASET

The Student Performance dataset was imported into Power BI using the **Get Data** option. The data was loaded into Power Query Editor for further inspection and transformation.

Procedure:

1. Open Power BI Desktop.
2. Select **Home** → **Get Data**.
3. Select the appropriate file format.
4. Choose the Student Performance dataset.
5. Select **Transform Data** to open Power Query Editor.

Dataset Imported into Power BI:



5. DATASET INSPECTION

The dataset was examined to understand its rows, columns, data types, and different categorical and numerical attributes.

Numerical Attributes

- Math Score
- Reading Score
- Writing Score

Categorical Attributes

- Gender
- Race/Ethnicity
- Parental Education
- Lunch
- Test Preparation Course

The data types of the columns were checked and corrected where required.

6. DATA CLEANING

The dataset was cleaned using Power Query Editor.

The following operations were performed:

- Checked for duplicate rows.
- Checked for missing or null values.
- Checked for errors.
- Verified appropriate data types.
- Removed unnecessary spaces.
- Standardized categorical values.
- Renamed columns where necessary.

The cleaned dataset was then reviewed to ensure that the data was consistent and ready for transformation.

Data Cleaning in Power Query:

The screenshot displays the Microsoft Power Query Editor interface. The main area shows a data table with the following columns: Student ID, gender, race/ethnicity, parental level of education, lunch, and test preparation course. Each column has a data type icon and a status indicator (Valid, Error, Empty) with a percentage. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Changed Type2'. The status bar at the bottom indicates '13 COLUMNS, 999+ ROWS'.

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	none
2	Female	Group C	Some College	Standard	completed
3	Female	Group B	Master's Degree	Standard	none
4	Male	Group A	Associate's Degree	Free/Reduced	none
5	Male	Group C	Some College	Standard	none
6	Female	Group B	Associate's Degree	Standard	none
7	Female	Group B	Some College	Standard	completed
8	Male	Group B	Some College	Free/Reduced	none
9	Male	Group D	High School	Free/Reduced	completed
10	Female	Group B	High School	Free/Reduced	none
11	Male	Group C	Associate's Degree	Standard	none
12	Male	Group D	Associate's Degree	Standard	none
13	Female	Group B	High School	Standard	none
14	Male	Group A	Some College	Standard	completed
15	Female	Group A	Master's Degree	Standard	none
16	Male	Group C	Some High School	Standard	none
17	Male	Group C	High School	Standard	none
18	Female	Group B	Some High School	Free/Reduced	none
19	Male	Group C	Master's Degree	Free/Reduced	completed
20	Female	Group C	Associate's Degree	Free/Reduced	none
21	Male	Group D	High School	Standard	none
22	Female	Group B	Some College	Free/Reduced	completed
23	Male	Group D	Some College	Standard	none
24	Female	Group C	Some High School	Standard	none
25	Male	Group D	Bachelor's Degree	Free/Reduced	completed
26	Male	Group A	Master's Degree	Free/Reduced	none
27	Male	Group B	Some College	Standard	none
28	Female	Group C	Bachelor's Degree	Standard	none
29	Male	Group C	High School	Standard	none
30	Female	Group D	Master's Degree	Standard	none
31	Female	Group D	Some College	Standard	none

Student ID were also created using Index column(from index 1):

The screenshot shows the Power Query Editor interface. The main area displays a table with the following columns: lunch, test preparation course, math score, reading score, writing score, and Student ID. The Student ID column is derived from the Index column (starting from 1). The table contains 31 rows of data. The right sidebar shows the 'Query Settings' pane with 'Properties' and 'Applied Steps' sections. The 'Applied Steps' section shows the sequence of transformations: Source, Promoted Headers, Changed Type, Added Index, Added Index1, and Removed Columns.

Index	lunch	test preparation course	math score	reading score	writing score	Student ID
1	standard	none	72	72	74	1
2	standard	completed	69	90	88	2
3	standard	none	90	95	93	3
4	free/reduced	none	47	57	44	4
5	standard	none	76	78	75	5
6	standard	none	71	83	78	6
7	standard	completed	88	95	92	7
8	free/reduced	none	40	43	39	8
9	free/reduced	completed	64	64	67	9
10	free/reduced	none	38	60	50	10
11	standard	none	58	54	52	11
12	standard	none	40	52	43	12
13	standard	none	65	81	73	13
14	standard	completed	78	72	70	14
15	standard	none	50	53	58	15
16	standard	none	69	75	78	16
17	standard	none	88	89	86	17
18	free/reduced	none	18	32	28	18
19	free/reduced	completed	46	42	46	19
20	free/reduced	none	54	58	61	20
21	standard	none	66	69	63	21
22	free/reduced	completed	65	75	70	22
23	standard	none	44	54	53	23
24	standard	none	69	73	73	24
25	free/reduced	completed	74	71	80	25
26	free/reduced	none	73	74	72	26
27	standard	none	69	54	55	27
28	standard	none	67	69	75	28
29	standard	none	70	70	65	29
30	standard	none	62	70	75	30
31	standard	none	69	74	74	31

7. CREATING TOTAL MARKS

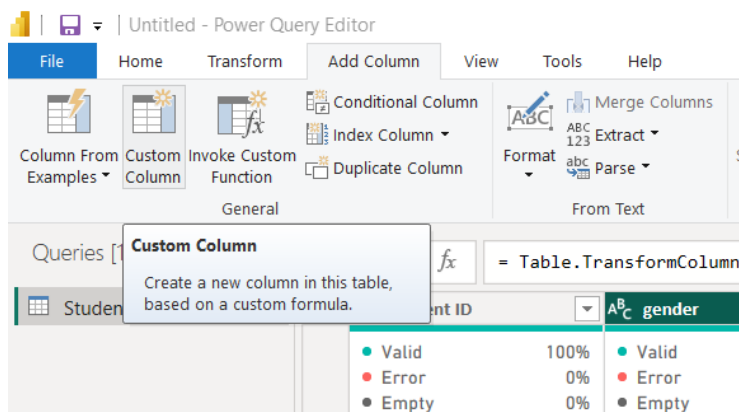
A **Total Marks** column was created by adding the Math, Reading, and Writing scores.

Power Query Formula

Total Marks = [Math Score] + [Reading Score] + [Writing Score]

Procedure:

1. Go to **Add Column**.
2. Select **Custom Column**.



3. Enter the formula.
4. Name the column **Total Marks**.
5. Click **OK**.

Total Marks Column:

Custom Column

Add a column that is computed from the other columns.

New column name
Total Marks

Custom column formula ⓘ
= [math score]+[reading score]+[writing score]

Available columns
gender
race/ethnicity
parental level of education
lunch
test preparation course
math score
reading score
writing score

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

1	Female				
2	Female				
3	Female				
4	Male				
5	Male				
6	Female				
7	Female				
8	Male				
9	Male				
10	Female				
11	Male				
12	Male				
13	Female				
14	Male				
15	Female				
16	Female				
17	Male				
18	Female				
19	Male				
20	Female				
21	Male				
22	Female	group B	some college	free/reduced	completed
23	Male	group D	some college	standard	none
24	Female	group C	some high school	standard	none
25	Male	group D	bachelor's degree	free/reduced	completed

8. CREATING AVERAGE SCORE

An **Average Score** column was created to calculate the student's average performance across the three subjects.

Formula

Average Score = [Total Score] / 3

This gives a single score representing the student's overall academic performance.

Average Score Column:

test preparation course	math score	reading score	writing score	Total Marks
completed	65	75	70	
none	44	54	53	
none	69	73	73	

Custom Column

Add a column that is computed from the other columns.

New column name
Average

Custom column formula ①
= [Total Marks]/3

Available columns
Student ID
gender
race/ethnicity
parental level of education
lunch
test preparation course
math score
reading score
writing score

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

9. CREATING PERFORMANCE GRADE:

A **Grade** column was created using conditional logic based on the Average Score.

The following classification was used:

Average Score Grade

80 – 100 A

60 – 79 B

40 – 59 C

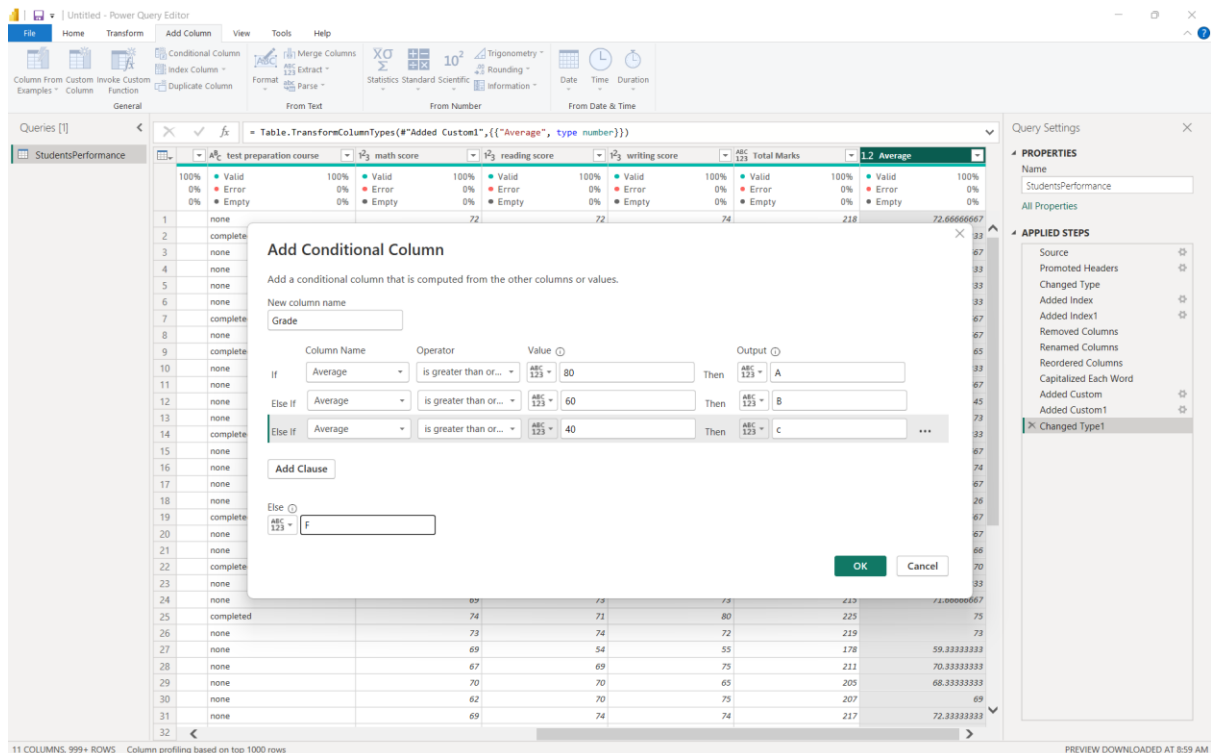
Below 40 F

Procedure

1. Select **Add Column**.
2. Select **Conditional Column**.
3. Select **Average**.

4. Add the required conditions.

5. Name the column **Grade**.



10. CREATING RESULT COLUMN

A **Result** column was created based on the student's Average Score.

The criterion used was:

- **Average Score ≥ 40 → Pass**
- **Average Score < 40 → Fail**

Procedure

1. Select **Add Column** → **Conditional Column**.
2. Select **Average Score**.
3. Set the condition as greater than or equal to 40.
4. Enter **Pass** for the true condition.
5. Enter **Fail** for the false condition.

Pass/Fail Classification:

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Result

Column Name: Average Operator: is greater than or... Value: 40 Output: Pass

Else: Fail

OK Cancel

Therefore, those four columns were created :

ABC 123 Total Marks	\$ Average	ABC Grade	ABC 123 Result
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
218	72.67	B	Pass
247	82.33	A	Pass
278	92.67	A	Pass
148	49.33	C	Pass
229	76.33	B	Pass
232	77.33	B	Pass
275	91.67	A	Pass
122	40.67	C	Pass
195	65.00	B	Pass
148	49.33	C	Pass
164	54.67	C	Pass
135	45.00	C	Pass
219	73.00	B	Pass
220	73.33	B	Pass
161	53.67	C	Pass
222	74.00	B	Pass
263	87.67	A	Pass
78	26.00	F	Fail
134	44.67	C	Pass
173	57.67	C	Pass
198	66.00	B	Pass
210	70.00	B	Pass
151	50.33	C	Pass
215	71.67	B	Pass
225	75.00	B	Pass
219	73.00	B	Pass
178	59.33	C	Pass
211	70.33	B	Pass
205	68.33	B	Pass
207	69.00	B	Pass
217	72.33	B	Pass

11. UNPIVOTING SUBJECT SCORES

The three subject columns:

- Math Score
- Reading Score
- Writing Score

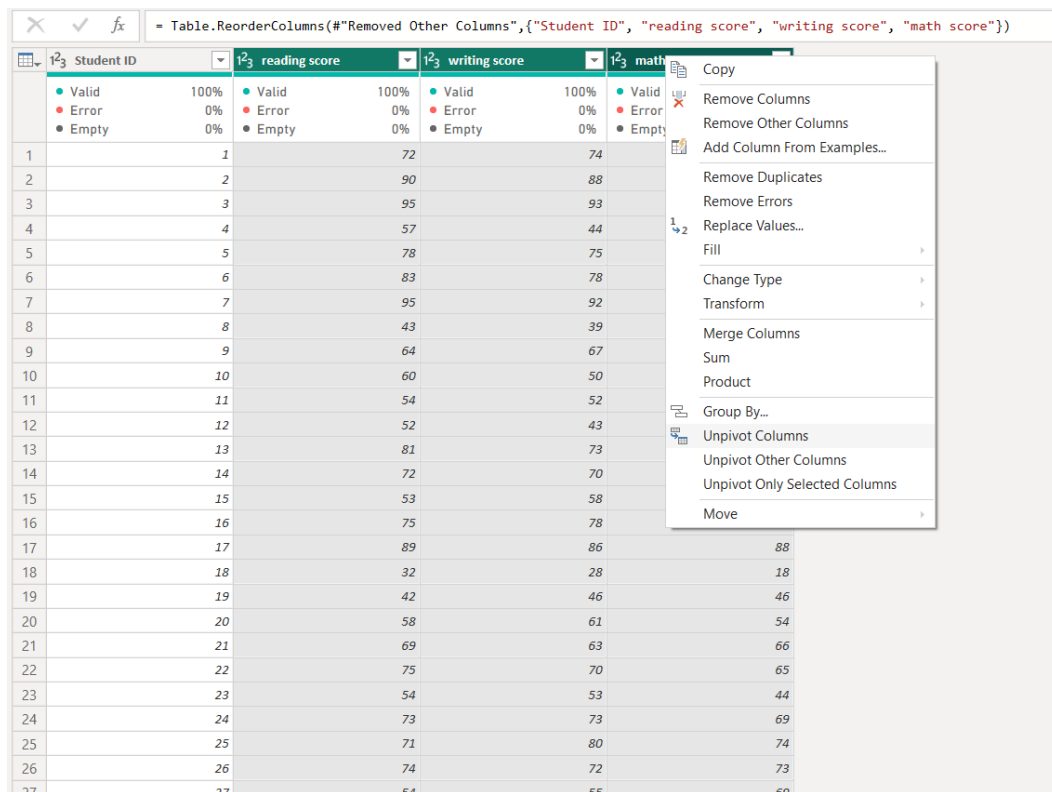
were transformed into a structured format using the **Unpivot Columns** option.

Procedure

1. Select the three subject score columns.
2. Right-click and select **Unpivot Columns**.
3. Rename **Attribute** as **Subject**.
4. Rename **Value** as **Score**.

This transformation makes it easier to compare student performance across subjects in Power BI visuals.

Unpivoted Subject Data:



The screenshot shows a Power BI Desktop interface with a table containing student scores. The table has four columns: Student ID, reading score, writing score, and math score. A right-click context menu is open over the 'math score' column, showing various options. The 'Unpivot Columns' option is highlighted. The formula bar at the top shows the table structure: Table.ReorderColumns(#"Removed Other Columns",{"Student ID", "reading score", "writing score", "math score"}).

Student ID	reading score	writing score	math score
1	72	74	
2	90	88	
3	95	93	
4	57	44	
5	78	75	
6	83	78	
7	95	92	
8	43	39	
9	64	67	
10	60	50	
11	54	52	
12	52	43	
13	81	73	
14	72	70	
15	53	58	
16	75	78	
17	89	86	88
18	32	28	18
19	42	46	46
20	58	61	54
21	69	63	66
22	75	70	65
23	54	53	44
24	73	73	69
25	71	80	74
26	74	72	73
27	54	55	69

= Table.RenameColumns(#"Unpivoted Columns",{{"Attribute", "Subject"}, {"Value", "Score"}})				Query Settings	
Student ID	Subject	Score		PROPERTIES	
Valid 100%	Valid 100%	Valid 100%		Name	
Error 0%	Error 0%	Error 0%		Student Scores	
Empty 0%	Empty 0%	Empty 0%		All Properties	
1	reading score	72		APPLIED STEPS	
2	writing score	74		Source	
3	math score	72		Promoted Headers	
4	reading score	90		Changed Type	
5	writing score	88		Added Index	
6	math score	69		Added Index1	
7	reading score	95		Removed Columns	
8	writing score	93		Renamed Columns	
9	math score	90		Reordered Columns	
10	reading score	57		Capitalized Each Word	
11	writing score	44		Added Custom	
12	math score	47		Added Custom1	
13	reading score	78		Changed Type1	
14	writing score	75		Added Conditional Column	
15	math score	76		Capitalized Each Word1	
16	reading score	83		Added Conditional Column1	
17	writing score	78		Removed Other Columns	
18	math score	71		Reordered Columns1	
19	reading score	95		Unpivoted Columns	
20	writing score	92		Renamed Columns1	
21	math score	88			
22	reading score	43			
23	writing score	39			
24	math score	40			
25	reading score	64			
26	writing score	67			
27	math score	64			
28	reading score	60			
29	writing score	50			
30	math score	38			
31	reading score	54			
32	writing score	52			

Then , close & apply.

Untitled - Power Query Editor

File Home Transform Add Column View Tools

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export res

Close & Apply Apply Close

StudentsPerformance Student Scores

1² Student ID

Valid 100% Error 0% Empty 0%

1 1 2 2 3 3 4 4

= Table.Ad

12. PREPARED DATA MODEL

After completing the transformations, the data model was reviewed to ensure that the required columns were available for analysis.

The prepared dataset contains:

- Student demographic information
- Subject scores
- Total Marks
- Average
- Grade
- Result
- Subject
- Score

13. CREATING DAX MEASURES

After applying the Power Query transformations, DAX measures were created for performance analysis.

Average Math score:

Average Math score = AVERAGE('Student Performance'[Math Score])

Average Reading score:

Average Reading score = AVERAGE('Student Performance'[Reading Score])

Average Writing score:

Average Writing = AVERAGE('Student Performance'[Writing Score])

Overall Average Score

Overall Average Score =

AVERAGE('Student Performance'[Average])

Overall Pass %

Pass % =

DIVIDE(

CALCULATE(

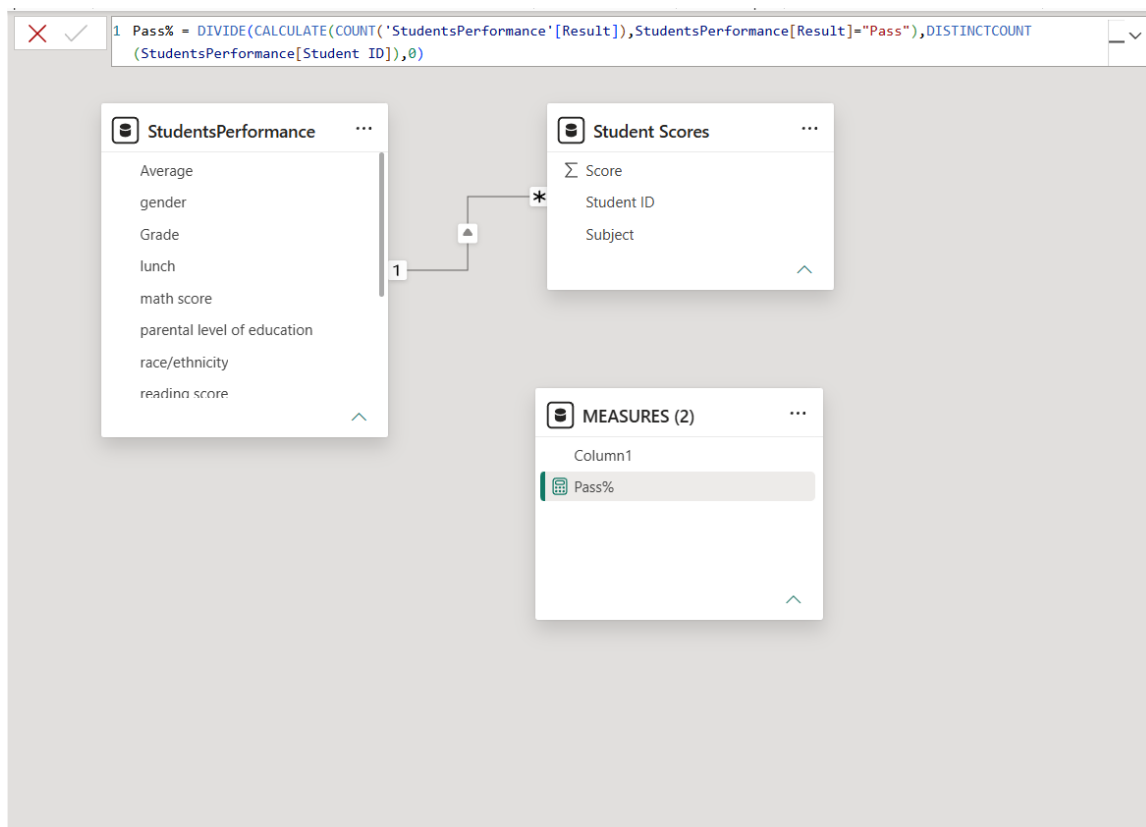
COUNT('Student Performance'[Result]),

'Student Performance'[Result] = "Pass"

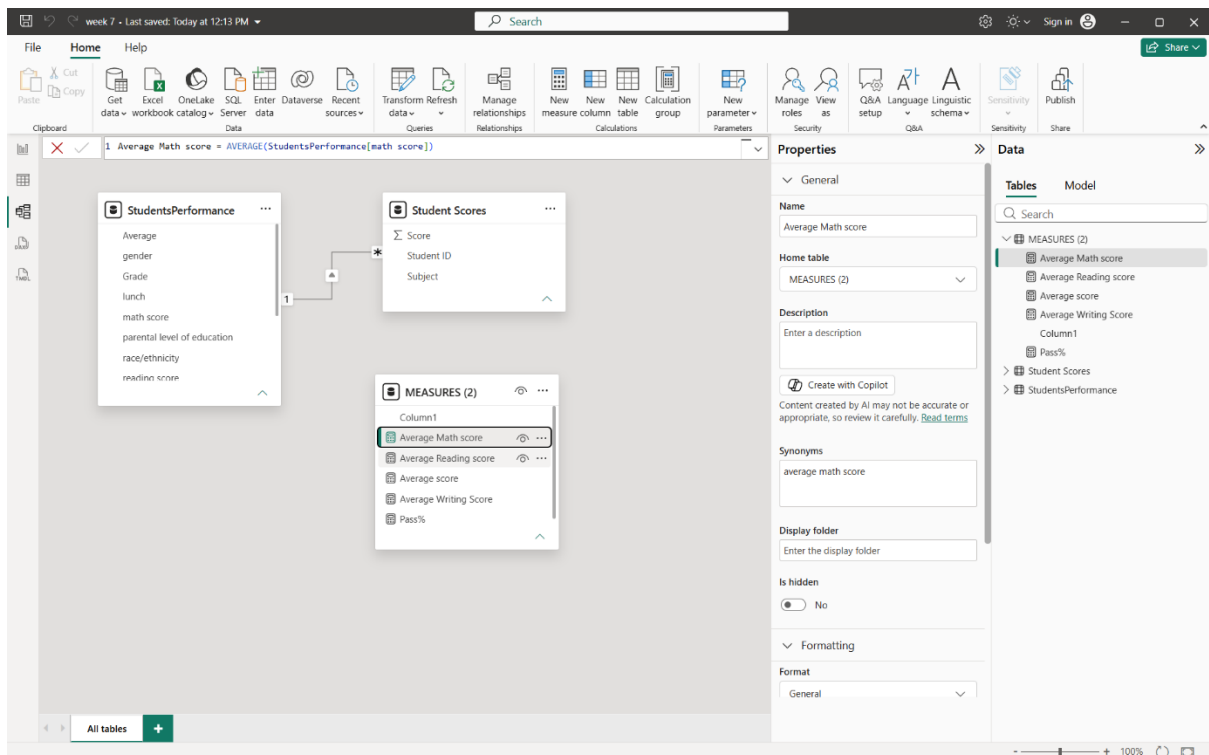
),

DISTINCTCOUNT('Student Performance'[Student ID]),0)

The **Overall Pass %** measure can be formatted as a percentage in Power BI.



Other DAX Measures:



14. VALIDATION

After completing the transformations and calculations, the dataset was validated.

The following checks were performed:

- Checked that there were no unwanted errors.
- Verified the calculated Total Score.
- Verified the Average Score.
- Checked the Performance Grade.
- Checked Pass/Fail classification.
- Verified the Subject and Score columns after unpivoting.
- Verified the DAX measures.
- Confirmed that the transformed data was ready for visualization.

15. RESULT

The Student Performance dataset was successfully cleaned and transformed using Power BI. New calculated columns such as **Total Marks, Average , Grade, and Result** were created. The subject scores were also unpivoted into **Subject** and **Score** columns.

The required DAX measures were created to support academic performance analysis. The prepared dataset can now be used to create an interactive **Student Performance Dashboard** in the next stage.

16. CONCLUSION

Week 7 activity provided practical experience in data cleaning, transformation, calculated columns, unpivoting, and DAX in Power BI. The final dataset is structured and ready for further analysis and dashboard development.